

test 3

Subject: Physics | Chapter: magnetism | Difficulty: hard

Question 1: Derive the expression for the magnetic field at a point on the axial line of a bar magnet of magnetic moment ' M ' at a distance ' r ' from the center of the magnet. Explain the conditions under which this expression simplifies to that of a magnetic dipole.

Question 2: A cyclotron accelerates protons to a kinetic energy of 10 MeV. If the magnetic field strength is 1.5 T, calculate the radius of the largest circular orbit the protons will follow. Explain the role of magnetic field in the cyclotron's operation and why it's essential for particle acceleration.

Question 3: A current-carrying loop in the shape of an equilateral triangle with side length ' a ' is placed in a uniform magnetic field B . Derive the expression for the maximum torque experienced by the loop when a current ' I ' flows through it. Explain how this torque can be used in a practical application like a DC motor.

Question 4: Explain the concept of magnetic hysteresis and its significance in choosing materials for transformers. Describe how hysteresis loss is related to the area of the hysteresis loop and its impact on transformer efficiency.

Question 5: Derive the expression for the magnetic field at the center of a circular coil of radius ' R ' carrying a current ' I '. Compare this field to that of a solenoid of the same radius and number of turns per unit length, discussing the differences and similarities in the field distribution.

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